

Analytical Study And Performance Evaluation of Kalka Dairy Farm

A Mid-Term Report For the BDM Capstone Project

Submitted by
Name : Yadvendra Pal
Roll Number : 24f1000137



IITM BS Degree Program
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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1. Executive Summary –

This project presents a data-driven study of **Kalka Dairy Farm**, focusing on improving business record management, operational efficiency, and decision-making. The study was developed around primary business information related to dairy animals, milk production, customer transactions, milk sales, feed consumption, veterinary activities, and operational expenses. The collected information was organized into structured datasets and analysed using descriptive statistics, trend analysis, cost and profitability analysis, and KPI-based reporting.

The dataset covers a **six-month period from January to June 2025** and includes operational and transactional records along with monthly summaries. Descriptive statistics were used to understand the central tendency, variation, minimum, maximum, median, and quartile values of important numerical variables. Visual analysis was then used to identify changes in milk production, revenue, expenses, and profitability over time.

The analysis indicates that milk production showed an overall improvement after a lower level in February and reached its highest level in May. Revenue and net profit generally followed the production trend, while profitability remained positive throughout the analysed period. The results also highlight the importance of monitoring feed consumption, veterinary expenses, operating costs, customer transactions, and sales quantities.

The project combines **Excel-based analysis, Pivot Tables, charts, KPIs, and Python-based statistical/visual analysis** to convert raw records into understandable business insights. The findings provide a practical basis for improving record accuracy, monitoring financial performance, identifying operational patterns, and supporting informed decisions at Kalka Dairy Farm.

2. Proof of Originality –

Primary Data Link- The Excel sheet link of Primary data which is collected from Kalka Dairy Farm for the analysis of the project is given below.

[Kalka_Dairy_Farm data.xlsx](#)

 Kalka_Dairy_Farm_BDM_Project.ipynb

Interaction with the Owner, video link- The google drive link of the in-person interaction meeting with the owner of kalka dairy farm Mr. Kalka Prashad.

 Interaction video.mp4

Letter from the Organisation- This is a letter from the business kalka dairy farm certifying the originality of the data and their permission to use their data for project purposes.

 Letter from organisation.pdf

Images of the kalka dairy farm-



(1) The photo with the owner Mr. kalka Prashad



(2) All dairy animals



(3) Animal milking



(4) Calf of Animals

3. Metadata –

1. Range of Date

Metadata Item	Details
Data Period	January 2025 to June 2025
Duration	6 months
Time Granularity	Daily records for milk production, sales and feed-related activities; monthly summary for business performance
Analysis Period	Six-month period used for descriptive statistics, visualization and business analysis

2. Data Sources

Metadata Item	Details
Primary Source	Kalka Dairy Farm business records and information associated with farm operations
Business Context	Records relating to dairy animals, milk production, customers, milk sales, feed consumption, veterinary activities and expenses
Data Format	Structured Excel workbook prepared for the BDM project
Supporting Source	Business-owner/domain discussions and information collected during the project data-collection process

3. Forms of Data

Metadata Item	Details
Structured Data	Tabular records maintained in Excel worksheets
Numerical Data	Milk quantities, sales quantity, selling rate, sales amount, feed quantity, feed cost, veterinary cost, expenses, animal age and purchase cost
Categorical Data	Animal/customer identifiers, customer types, expense categories and other descriptive fields
Date Data	Daily dates used for time-based analysis
Summary Data	Monthly production, revenue, expenses, profit and profit-margin measures

4. Data Collection Process

1. Business problem identification – understand the operational and record-management problems faced by the dairy farm.
2. Requirement identification – identify the information needed for animals, milk production, customers, sales, feed, veterinary activities and expenses.
3. Data collection – collect available business records for the selected six-month period.
4. Data organization – arrange the collected information into separate Excel worksheets.
5. Data cleaning and preparation – check formats, missing/invalid values and numerical fields before analysis.
6. Data analysis – calculate descriptive statistics and prepare charts, KPIs and business insights.

5. Dataset / Sheet-wise Metadata

Dataset 1: Animal_Master

Purpose: Stores master information about dairy animals. Records/rows: 10.

Attributes: Animal ID; Tag No; Animal Name; Type; Breed; Gender; DOB; Age (Years); Purchase Date; Purchase Cost (₹); Status; Avg Milk/Day (L)

Dataset 2: Daily_Milk_Production

Purpose: Records daily milk production for each animal. Records/rows: 1,448.

Attributes: Date; Animal ID; Animal Name; Morning (L); Evening (L); Total Milk (L); Remarks

Dataset 3: Customers

Purpose: Stores customer information and milk purchasing details. Records/rows: 40.

Attributes: Customer ID; Customer Name; Village; Mobile No.; Customer Type; Quantity/Day (L); Price/Litre (₹); Payment Mode; Status

Dataset 4: Milk_Sales

Purpose: Stores daily milk sales transactions. Records/rows: 1,086.

Attributes: Sale ID; Date; Customer ID; Customer Name; Milk (L); Rate; Amount; Payment Mode; Payment Status

Dataset 5: Feed_Consumption

Purpose: Tracks daily feed usage and feed costs. Records/rows: 1,448.

Attributes: Record ID; Date; Animal ID; Animal Name; Feed Type; Quantity (Kg); Cost per Kg (₹); Total Cost (₹); Supplier

Dataset 6: Veterinary_Records

Purpose: Records animal health check-ups and treatments. Records/rows: 74.

Attributes: Record ID; Date; Animal ID; Animal Name; Treatment Type; Veterinarian; Medicine; Cost (₹); Health Status; Next Visit

Dataset 7: Expenses

Purpose: Maintains operational expense records. Records/rows: 724.

Attributes: Expense ID; Date; Category; Description; Amount (₹); Payment Mode; Vendor

Dataset 8: Monthly_Summary

Purpose: Provides monthly business performance indicators. Records/rows: 6.

Attributes: Month; Milk Produced (L); Revenue (₹); Feed Cost (₹); Other Expenses (₹); Total Expenses (₹); Net Profit (₹); Avg Daily Milk (L); Profit Margin (%)

Dataset 9: Dashboard_Data

Purpose: Contains data prepared for dashboard visualization and reporting. Records/rows: 40.

Attributes: Monthly Milk Production

Dataset 10: KPI

Purpose: Defines key performance indicators for the dairy farm. Records/rows: 7.

Attributes: KPI; Value

6. Metadata Interpretation

- The dataset is structured and primarily tabular, making it suitable for Excel Pivot Tables, charts and Python/Pandas analysis.
- The operational sheets capture the main business activities: livestock, milk production, customers, sales, feed, veterinary care and expenses.
- Monthly_Summary, Dashboard_Data and KPI/KPI_Metadata support business-level interpretation rather than raw transaction entry.
- The metadata links the collected information to the project's business objective of improving record management, operational efficiency and decision-making.

7. Relevance to the Problem Statement

The proposal identifies the need for a centralized business data management system for accurate records of dairy animals, milk production, customer transactions, feed consumption and operational expenses. The workbook structure directly represents these areas and therefore provides the data foundation for financial monitoring, reduction of record-keeping errors and analysis of farm performance.

8. Overall Dataset Metadata

The uploaded workbook contains 10 worksheets and approximately 4,883 non-empty data rows across the sheets. It combines operational, transactional and summary data. The dataset is designed to support descriptive analysis, trend analysis, cost analysis, financial performance analysis and KPI/dashboard reporting.

4. Descriptive Statistics –

Descriptive statistics summarize the six-month dataset (January 2025–June 2025) include Count, Mean, 25th Percentile (Q1), 50th Percentile (Median/Q2), 75th Percentile (Q3), Standard Deviation, Minimum and Maximum.

1. Animal Master2. Daily Milk Production

Variable	Count	Mean	25%	50%	75%	Std. Dev.	Min	Max
Morning Milk (L)	1448	5.60	5.10	5.70	6.10	0.61	4.30	6.70
Evening Milk (L)	1448	5.17	4.70	5.20	5.60	0.57	3.90	6.20
Total Milk (L)	1448	10.77	9.80	10.90	11.70	1.18	8.20	12.90

3. Customers

Variable	Count	Mean	25%	50%	75%	Std. Dev.	Min	Max
Quantity/Day (L)	40	3.16	1.50	2.50	4.00	2.44	1.50	10.00
Price/Litre (₹)	40	59.70	59.00	60.00	60.00	0.72	58.00	60.00

4. Milk Sales

Variable	Count	Mean	25%	50%	75%	Std. Dev.	Min	Max
Milk Sold (L)	1086	4.51	2.00	3.00	6.00	2.97	1.50	10.00
Rate (₹/L)	1086	59.46	59.00	60.00	60.00	0.89	58.00	60.00
Amount (₹)	1086	265.79	120.00	180.00	360.00	170.84	90.00	580.00

5. Feed Consumption

Variable	Count	Mean	25%	50%	75%	Std. Dev.	Min	Max
Quantity (Kg)	1448	13.08	8.00	15.00	18.00	5.25	6.00	20.00
Cost per Kg (₹)	1448	14.75	7.00	8.00	25.00	10.34	6.00	28.00
Total Cost (₹)	1448	141.75	96.00	120.00	180.00	43.89	96.00	224.00

6. Veterinary Records

Variable	Count	Mean	25%	50%	75%	Std. Dev.	Min	Max
Treatment Cost (₹)	74	629.05	500.00	600.00	750.00	172.21	350.00	850.00

7. Operational Expenses

Variable	Count	Mean	25%	50%	75%	Std. Dev.	Min	Max
Expense Amount (₹)	724	982.75	200.00	448.00	1,400.00	1,273.89	55.00	4,979.00

8. Monthly Business Performance

Variable	Count	Mean	25%	50%	75%	Std. Dev.	Min	Max
Milk Produced (L)	6	1,155.00	1,120.00	1,165.00	1,185.00	41.87	1,088	1,201
Revenue (₹)	6	69,300.00	67,800.00	69,900.00	71,100.00	2,511.99	65,280	72,060
Feed Cost (₹)	6	18,886.67	18,300.00	18,985.00	19,400.00	616.82	17,900	19,600
Other Expenses (₹)	6	14,500.00	14,100.00	14,560.00	14,800.00	374.11	13,950	14,980
Total Expenses (₹)	6	33,386.67	32,400.00	33,610.00	34,200.00	983.07	31,850	34,580
Net Profit (₹)	6	35,913.33	35,400.00	36,290.00	36,800.00	1,537.90	33,430	37,480
Avg Daily Milk (L)	6	38.50	37.50	38.85	39.40	1.38	36.30	40.00
Profit Margin (%)	6	51.81	51.50	51.92	52.10	0.37	51.21	52.26

Key Findings

- The dataset contains records for 10 animals and 40 customers over six months.
- Average total milk production per animal-day record is 10.77 litres.
- There are 1,086 milk-sales transactions, with an average transaction value of ₹265.79.
- Average monthly revenue is ₹69,300 and average monthly expenses are ₹33,386.67.
- Average monthly net profit is ₹35,913.33, with an average profit margin of 51.81%.
- Expense values show considerable variation, reflected by the relatively high standard deviation.

I am also providing the colab notebook for reference of descriptive statistics.

[Kalka_Dairy_Farm_BDM_Project.ipynb](#)

5. Detailed Explanation of Analysis Process –

1. Objective of the Analysis

The main objective of the analysis is to convert the dairy farm's organized business records into actionable information for better operational efficiency and decision-making. The project focuses on animal records, daily milk production, customer transactions,

milk sales, feed consumption, veterinary records and operational expenses. The analysis is therefore not limited to calculating numbers; it connects production, demand, costs and financial performance with the two identified problem statements: maintaining accurate and organized records and improving financial monitoring and farm performance.

2. Analysis Process

Process / Method	Application in the Project
Data validation and preparation	Check the Excel records for missing values, duplicate entries, inconsistent dates, categories and incorrect numerical formats. Standardized data provides a reliable base for analysis.
Data organization and integration	Use the separate worksheets for Animal_Master, Daily_Milk_Production, Customers, Milk_Sales, Feed_Consumption, Veterinary_Records, Expenses, Monthly_Summary, Dashboard_Data and KPI information as related business components.
Descriptive analysis	Calculate count, total, mean, median, minimum, maximum, range and quartiles such as 25%, 50% and 75% for appropriate numerical variables.
Trend and comparative analysis	Compare daily/monthly milk production, sales revenue, expenses, feed cost and profit to identify changes, fluctuations and important periods.
Cost and profitability analysis	Break down operating costs and compare revenue with total expenses to understand net profit and profit margin.
Visualization and dashboard analysis	Use charts, Pivot Tables and KPI summaries to make patterns easier for the business owner to understand.
Business interpretation	Translate the numerical results into findings, identify possible causes of observed patterns and develop practical recommendations.

3. Methods Selected for the Project

- Descriptive Statistics – summarizes the large operational dataset and shows the central tendency, spread and distribution of important numerical variables.
- Trend Analysis – examines how production, sales, expenses and profit change over the six-month period and helps identify recurring increases or decreases.
- Cost and Profitability Analysis – identifies major expense areas and evaluates revenue, total expenses, net profit and profit margin.
- Customer and Sales Analysis – examines milk quantity sold, customer activity, payment modes and sales values to understand demand and transaction patterns.
- KPI and Dashboard Analysis – combines the most important indicators into a concise management view for quick monitoring.

4. Why These Methods Are Appropriate

These methods are appropriate because the project is a business data management study rather than a purely theoretical statistical exercise. The available workbook contains structured transactional and operational records over a six-month period. Descriptive statistics provide the required baseline, trend analysis explains time-based movement, cost/profitability analysis addresses the financial objective, and customer/sales analysis addresses the transaction side of the business. KPI and

dashboard analysis then converts the results into a form that a business owner can use for decision-making.

Method Justification and Comparison

5. Why This Method Is More Appropriate Than the Alternatives

Method Used	Why It Fits This Project	Why Other Methods Alone Are Less Suitable
Descriptive Statistics	The farm needs a clear summary of production, sales, costs and other records before decisions are made.	A complex predictive model would not first explain the basic current performance and distribution of the records.
Trend Analysis	The records are time-based and cover six months, so monthly/daily comparisons can reveal changes and fluctuations.	A single overall average would hide month-to-month changes and important periods.
Cost & Profitability Analysis	The problem statement specifically emphasizes financial monitoring and farm performance.	Only analysing production would not explain whether increased production is financially beneficial.
Customer & Sales Analysis	Milk sales and customer transactions are central to understanding demand and revenue.	Animal-level analysis alone cannot explain customer demand or sales performance.
KPI / Dashboard	The business owner needs a simple management view of important indicators.	Raw tables or lengthy statistical output are harder to use for routine decision-making.

6. Tools and Techniques Used

Microsoft Excel: Used for data organization, formulas, Pivot Tables, descriptive summaries, charts and dashboard preparation. It is appropriate because the project data is already maintained in structured Excel worksheets and the outputs are easy for a business owner to review.

Python / Pandas: Used where programmatic cleaning, grouping, aggregation and statistical calculations are required. It improves repeatability and reduces manual calculation errors.

Matplotlib: Used to create clear visual representations such as line charts and bar charts for trends and comparisons.

7. Expected Analytical Outputs

- Descriptive statistics tables for key numerical variables, including mean, median and 25%, 50% and 75% quartiles where relevant.
- Monthly/daily production and sales trend charts.
- Expense and feed-cost breakdowns.
- Revenue, total expense, net profit and profit-margin calculations.
- Customer and sales summaries.
- KPI cards and a management dashboard.
- Findings and recommendations directly connected to the identified business problems.

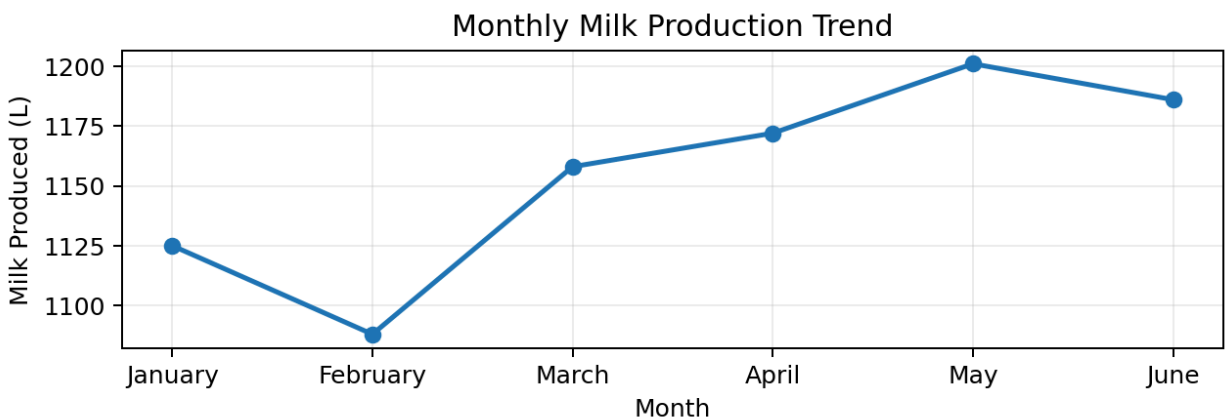
8. Link with the Problem Statement

The selected approach directly supports the project's two problem statements. First, structured data preparation and analysis provide a centralized and organized view of animal, production, customer, feed and expense records. Second, descriptive, trend, cost, profitability, sales and KPI analysis strengthens financial monitoring and decision-making. Therefore, the chosen methods are practical, understandable and directly aligned with the business objectives of Kalka Dairy Farm.

6. Results and Findings –

Analysis basis: The findings below are based on the Monthly_Summary and descriptive-statistics outputs from the Kalka Dairy Farm Colab analysis. The monthly data contains milk production, revenue, feed cost, other expenses, total expenses, net profit, average daily milk and profit margin.

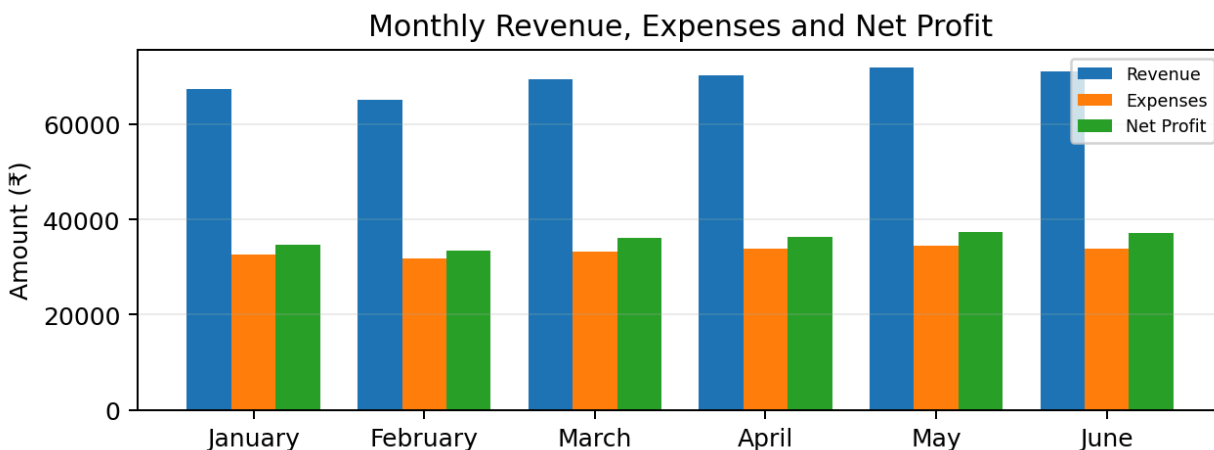
1. Milk Production Trend



Finding: Milk production started at 1,125 L in January, fell to the six-month low of 1,088 L in February, and then increased through May, when it reached the highest level of 1,201 L. Production reduced slightly in June to 1,186 L. Overall, January-to-June production increased by 61 L (5.4%).

Interpretation: The pattern shows a temporary early-period decline followed by sustained improvement. The relatively small month-to-month variation suggests that production remained broadly stable while the later months performed better. The descriptive analysis also reports an average monthly production of 1,155 L and an average daily production of 38.5 L.

2. Revenue, Expenses and Profit



Finding: Revenue moved from ₹67,500 in January to a peak of ₹72,060 in May, while total expenses rose from ₹32,700 to ₹34,580. Net profit followed the revenue pattern, increasing from ₹34,800 to a peak of ₹37,480 in May before a small decline in June.

Interpretation: The parallel movement of revenue and profit indicates that stronger production months were associated with higher financial returns. Expenses also increased as activity increased, but they remained below revenue in each monthly observation.

Results and Findings – Interpretation

3. Profitability Pattern

Month	Revenue (₹)	Net Profit (₹)	Profit Margin (%)
January	67,500	34,800	51.56%
February	65,280	33,430	51.21%
March	69,480	36,130	52.00%
April	70,320	36,450	51.83%
May	72,060	37,480	52.01%
June	71,160	37,190	52.26%

Finding: Profit remained positive in all six months, ranging from ₹33,430 to ₹37,480. The average monthly profit was ₹35,913, with an average profit margin of 51.81%. This shows stable profitability.

4. Sales and Customer Findings

- 1,086 milk-sale transactions were recorded.
- Average milk sold was 4.51 L per transaction.
- Average transaction value was ₹265.79.
- Selling price remained mostly stable around ₹60/L.
- Records include 40 customers and 10 animals.

5. Feed, Veterinary and Expense Findings

- Average feed consumption: 13.08 kg per record.
- Average feed cost: ₹141.75.
- Average veterinary cost: ₹629.05.
- Expenses showed considerable variation, highlighting the need for regular cost monitoring.

6. Overall Business Interpretation

The farm showed an overall improving trend after February. May recorded the highest milk production, revenue and profit, while June remained strong with a slight decline.

7. Key Findings

- Production improved after February.
- Revenue and profit followed the production trend.
- Profitability remained stable.
- Milk prices were relatively consistent.
- Regular expense and KPI monitoring can improve decision-making.

8. Conclusion

The analysis shows **stable farm performance with improving production and financial returns** . Regular monitoring of milk production, sales, feed costs, and expenses can support **better decisions and improved farm performance**.